

THE MAMBA-BASED SYSTEM FOR DCASE 2025 SOUND EVENT LOCALIZATION AND DETECTION CHALLENGE

Technical Report

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ABSTRACT

This technical report gives an overview of our system for task 3 in DCASE 2025 challenge. We propose a stereo sound event localization and detection (SELD) system using the mel spectral and intensity vector as the input features. We construct a Mamba-based network and achieve the significant improvements compared to the baseline system with few data augmentation methods. We conduct the performance evaluation on the dev-test of the Stereo SELD dataset extracted from the Sony-TAU Realistic Spatial Soundscapes 2023 (STARSS23) dataset.

Index Terms— Stereo sound event localization and detection, source distance estimation, Mamba, audio-visual fusion

1. INTRODUCTION

Sound event localization and detection (SELD) involves determining the position and class of the active sound sources over time simultaneously. It has broad applications, such as human-robot interaction, tele-conferencing, and security. In the previous year's challenge, the task involved estimating the distance of the active sound events in addition to their positions and classes. The dataset includes 360° video recordings and audio data. Two types of formats can be used for the audio data: first-order ambisonics (FOA), recorded with 32 microphones, and 4-channel recordings made with a tetrahedral microphone array (MIC). Unlike the previous challenges, this year's challenge only includes stereo audio data.

In this challenge task, most studies utilized the ResNet-Conformer architecture to achieved good results. However, Mamba has been widely used in recent years for various speech-related tasks, including speech enhancement, speech separation, and audio classification [1], [2]. Therefore, In this report, we explore how to leverage Mamba blocks to enhance the performance of audio-visual SELD task.

Similar to the ResNet-Conformer architecture developed previously, we replaced the Conformer with the Mamba module, named ResNet-Mamba. The results demonstrate that the proposed system performs well in this task.

2. METHOD

2.1. Data augmentation

The official dataset, DCASE2025 Task3 Stereo SELD Dataset, contains 41.7 hours of real recordings sampled from the original STARSS23 recordings. To further improve the diversity and amount of the training dataset, the data augmentation technique is employed.

The method is to simulate the new stereo data. We first generate the First-of-Order (FOA) audio data and then transform the FOA audio data to the stereo data. Specifically, the single-channel audio data, selected from the FSD50K dataset [3] are employed to convolve with the Spatial Room Impulse Responses (SRIRs) by leveraging a tool called SpatialScaper [4]. The duration of generated audio clips is 5 s. Finally, we synthesized 125 hours of data.

2.2. The architecture of the neural network

The overall framework of the proposed audio-visual sound event detection and localization framework using the Mamba blocks is illustrated in Fig. 1. The proposed neural network consists of two feature encoders, namely audio encoder and visual encoder. In this challenge, the audio encoder is composed of ResNet-18 [5] and two Mamba Blocks, as shown in Fig. 2. Each Mamba block includes 8 Bi-mambas [6], respectively. For the visual encoder, we employ the ResNet-50 pretrained on the ImageNet dataset to extract the visual features.

As described in Fig. 1, stereo audio data is utilized. Instead of directly inputting the stereo audio signals into the system, spectrograms are generated first using the Short-Time Fourier Transform (STFT) with a 512-point Hanning window and a hop length of 150 samples. Subsequently, the 5-second audio signals sampled at 24 kHz are discretized into 800 temporal bins. Additionally, the intensity vector of the stereo audio data is computed. The 2-channel mel spectral features and the one-channel intensity vector are concatenated to form 3-channel features. To align with the official dataset, the input segment length of the neural network is set to 5 seconds. The shape of the input audio feature is $3 \times 800 \times 128$. Similarly, the shape of the input images is $3 \times 224 \times 224$.

After the encoder achieves the feature extraction, we obtain audio and visual features with dimensions of 50×512 , respectively. These extracted features are then feed into the fusion block, which consists of two components: the Mamba attention module and the Bi-Mamba module. The Mamba attention module, a variation of the Bi-Mamba module. We concatenate the audio and visual features

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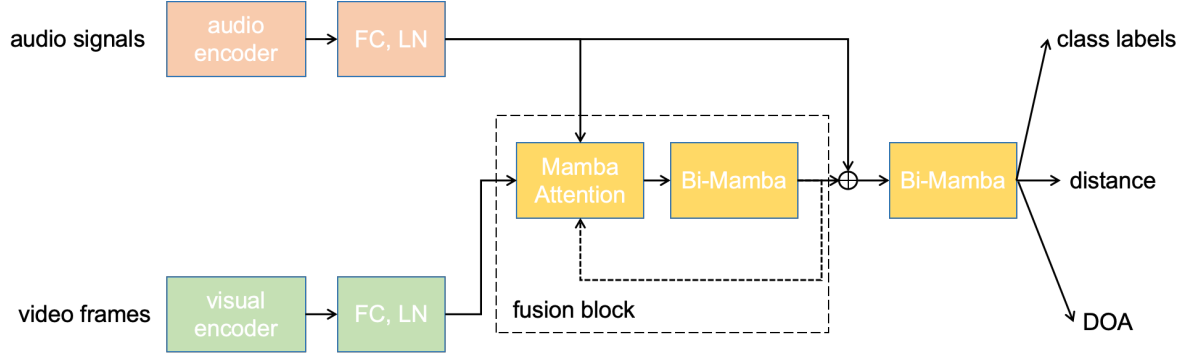


Figure 1: Overview of our SELD system.

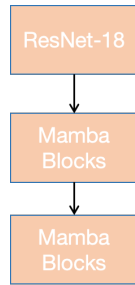


Figure 2: The architecture of audio encoder.

along the temporal dimension within the Mamba attention module. To fully exploit the complementary nature of audio and visual information, the concatenated fused features are flipped. Subsequently, both the original and flipped fused features are fed into the Mamba attention module simultaneously. The augmented visual features are separated from the fused features along the time dimension. The subsequent Bi-Mamba block is utilized to further refine the features. In this challenge, three fusion blocks are employed to integrate audio and visual information. Notably, the audio feature remains constant across all fusion modules, while the input visual feature consists of the augmented visual features generated from the preceding fusion module. Unlike the audio feature encoder, the Mamba Block within the fusion module contains only four Mambas. Following the data fusion process, the audio and visual features are concatenated along the feature dimension and passed through a Bi-Mamba block. Finally, we can obtain the expected output following the multi ACCDOA format [7].

Additionally, the audio encoder is pretrained using simulated data. During training on the challenge data, the parameters of the visual encoder are kept constant, and only the audio encoder is trained.

3. EXPERIMENTS AND RESULTS

We present the results on the Stereo dev-test split for the audio-visual system in Table. 1, where we compare our approach with the challenge baseline using the performance metrics specified in the challenge.

Notably, our system present large improvements in all the performance metrics compared to the baseline system.

Table 1: Macro-averaged test results on Stereo SELD dataset.

Method	$F_{20^\circ} \uparrow$	$F_{20^\circ/on} \uparrow$	DOAE $\downarrow$	RDE $\downarrow$	OSA $\uparrow$
Baseline-AV	26.8%	20.0%	23.8 $^\circ$	40%	80%
Ours	32.4%	24.1%	18.1°	32.88%	81.44%

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